The Mathematics of `width: 100%`

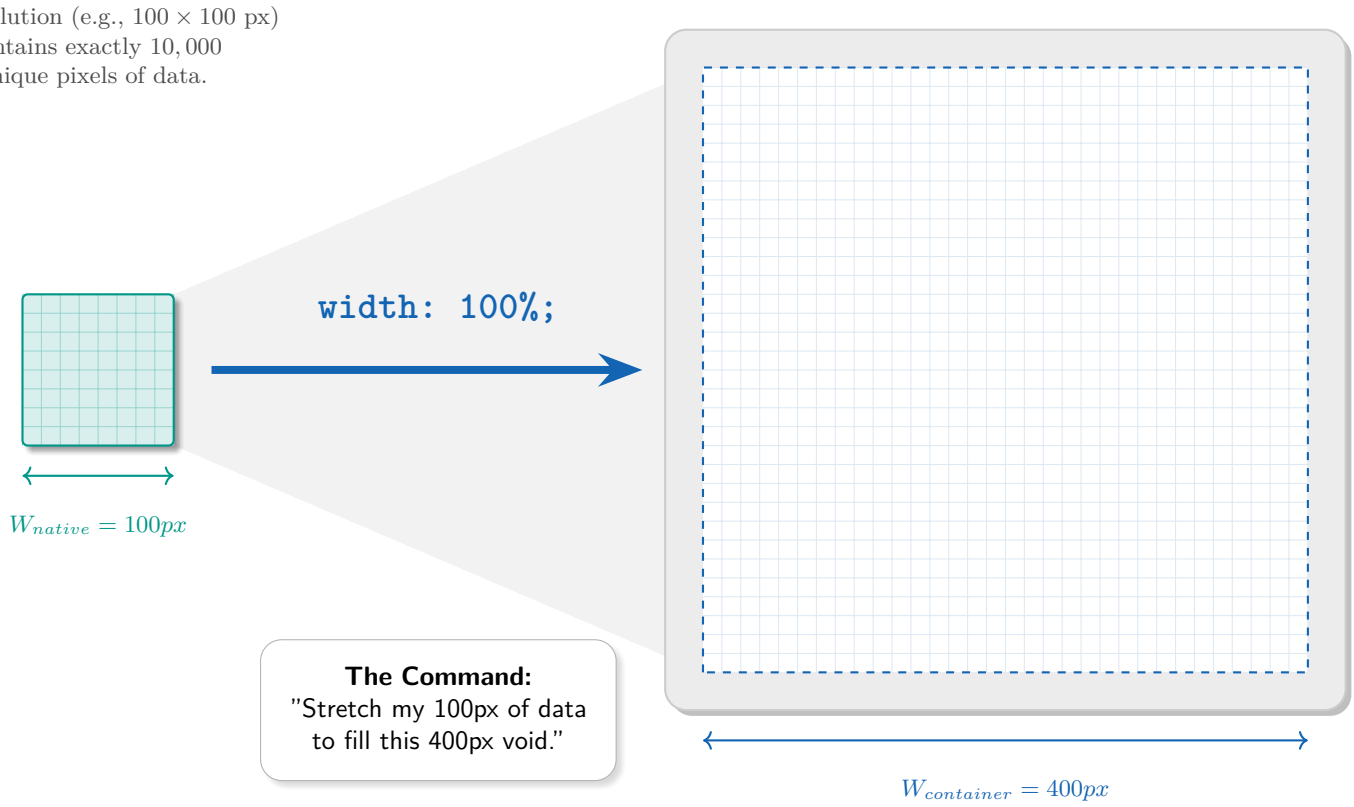
Why scaling low-resolution images forces extreme pixelation

Native Image File

Low resolution (e.g., 100×100 px)
Contains exactly $10,000$
unique pixels of data.

Web Container

High-density Display (e.g., 400×400 px)
Requires $160,000$ pixels to fill the space.



The Mathematical Deficit

1. **Scale Factor (1D):** $S = \frac{W_{container}}{W_{native}} = \frac{400px}{100px} = 4$
2. **Area Multiplier (2D):** Since images are 2D grids, the total pixel requirement scales quadratically: $Area = S^2 = 4^2 = \mathbf{16}$
3. **The Gap:** The browser needs $160,000$ pixels, but only has $10,000$ real pixels. It must **invent 15 fake pixels** for every 1 real pixel.

How Browsers "Invent" Pixels (Interpolation Algorithms)